CHAIN RESTAURANT CALORIE POSTING LAWS, OBESITY, AND CONSUMER WELFARE

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Abstract
This paper investigates whether and why laws requiring chain restaurants to post calories on menus and menu boards work. We develop a model of calories consumed that highlights multiple potential channels through which these laws influence choice and that outlines an empirical strategy to disentangle these alternatives. We test the predictions of our model using data from the Behavioral Risk Factor Surveillance System on body mass index and consumer well-being, as well as our own surveys on how the law influences where people eat and how randomized exposure to calorie information affects feelings toward menu items. Viewed in its totality, our results are consistent with an economic model in which calorie labels influence consumers both by providing salient information and by imposing a welfare-reducing moral cost (or feelings of guilt) on unhealthy eating. (JEL: I12, I18, I31, D91)

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Teaching Slides

A set of Teaching Slides to accompany this article is available online as Supplementary Data.

1. Introduction

There is a large literature exploring the causal link between the consumption of meals at restaurants and the rise in obesity (e.g. Currie et al. 2010; Dunn 2010; Anderson and Matsa 2011; Courtemanche et al. 2016). From a policy perspective, this connection raises the question of whether consumers are inadequately informed about nutritional content—a concern enhanced by evidence that individuals tend to underestimate calories in restaurant meals (e.g. Wisdom et al. 2010; Block et al. 2013). Relatedly, nutrition could be a shrouded attribute about which consumers are inattentive at the time of purchase, even if they are well-informed. To address such internalities, policymakers have introduced mandates that require chain restaurants to post calorie counts on menus and menu boards—a "nudge" designed to increase information and salience at the point of purchase.

Prior literature has focused on estimating the impacts of such mandates on either calories consumed (e.g. Elbel et al. 2009, 2011; Bollinger et al. 2011; Finkelstein et al. 2011) or body mass index (BMI) (e.g. Deb and Vargas 2016; Restrepo 2017). Yet, the literature provides little insight as to the channels through which such "nudges" influence choice or the associated welfare effects, a gap that our study aims to fill. In doing so, we seek to move the literature from evaluating whether a given "nudge" works to understanding whether and why it works.

We begin by developing a model of calories consumed and how this choice may be impacted by calorie labels. We posit three channels through which calorie labels may influence choice: (i) beliefs about the caloric content of a meal, (ii) other "internalities" or biases such as inattention, and (iii) moral costs or feelings of guilt. Moral (or

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psychological) costs depend on the environment and reflect the notion that individuals may feel guilty when selecting high calorie items (Choi et al. 2019; Kim 2020).[1]

We use the model to elucidate how calorie labels could influence each of these channels and derive the subsequent impact on calories consumed and welfare. First, providing calories on menus can improve the accuracy of individuals' beliefs about caloric content. Second, calorie mandates increase the salience of healthy food options at the time of purchase, which can reduce non-information internalities. Third, such labels may impose a "shadow tax" on higher calorie items (due to moral costs), which increases their "price" relative to lower calorie alternatives. We show that while all three channels lead to reductions in calories consumed, they have differential impacts on welfare. Whereas changes in calories consumed that arise through reductions in informational and non-informational internalities are welfare enhancing, those that arise through increased moral costs lead to reductions in welfare.

We use our model to formulate empirical strategies to distinguish between these channels. Our primary analysis utilizes data from the 1994–2012 waves of the Behavioral Risk Factor Surveillance System (BRFSS). We supplement these with data from two primary surveys that explore how calorie labels impact (i) the likelihood of eating at chain and non-chain restaurants and (ii) perceptions of social appropriateness and guilt when ordering menu items with different calorie amounts. Rather than relying on a single test to identify channels, our approach examines impacts across multiple outcomes and for different subsets of the population.

We estimate the impact of local calorie mandates on BMI using the BRFSS data. Identifying variation comes from differences in implementation and enforcement across geographic areas and over time. We find that calorie mandates lead to a small but statistically significant average reduction in BMI of 0.17 kg/m^2 (approximately 1.5 pounds at the average height or 0.7% of the sample mean), with some evidence suggesting a more pronounced effect in areas with active enforcement. We provide evidence from event-study models and other robustness checks that the BRFSS results are causal and are not driven by policy endogeneity, the construction of the control group, the limited number of policy changes, confounding by the Great Recession, or bias from treatment effect heterogeneity and staggered policy adoption (Goodman-Bacon 2021). We provide further evidence to rule out compensating behaviors such as smoking, drinking, or exercise.

We next explore mechanisms through which calorie mandates influence BMI. We find evidence consistent with calorie labels impacting choice due to both the provision (or salience) of information and moral costs. Evidence of moral costs comes from four distinct sources. First, BRFSS analyses show that calorie labeling mandates lead to a *reduction* in self-reported life satisfaction that is concentrated among people in the healthy weight category, who presumably benefit less than others from information and salience, enabling moral costs to dominate and turn the net effect on welfare negative.

1. While guilt seems to decrease if consumption is delayed (Duke and Amir 2019), in the case of restaurant food consumption, there is very little delay between the purchase and the consumption of the meal.

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Second, we estimate quantile regressions for BMI that allow heterogeneous responses across the BMI distribution and show that labeling laws have statistically significant impacts amongst healthy weight individuals, whose choices are the least likely to be affected by internalities. Third, a non-trivial fraction of survey respondents indicate that they would reduce the frequency with which they eat at restaurants posting calories following the implementation of the national mandate. This suggests a willingness to eat at a less preferred restaurant to avoid the guilt associated with dining in an environment with calorie labels. Finally, our survey experiment shows that randomized exposure to calorie information influences proxies for moral costs—feelings of guilt and the perceived social appropriateness of ordering a given menu item.

Evidence on the importance of information and/or its salience comes from two distinct sources. First, results from the quantile regressions show that the largest point estimates come from the 0.9 quantile—the group for whom internalities, and hence informational effects, are likely most pronounced. Second, more than a quarter of survey respondents state that the introduction of calorie labels at the national level would lead them to increase the frequency with which they eat at chain restaurants. This suggests that these consumers believe that calorie labels provide valuable information that leads to greater utility when eating at such restaurants.

Our paper contributes to several literatures. The first explores how simple messages that provide information about misperceived or shrouded attributes impact subsequent choices. For example, a growing body of work explores how messages that educate individuals about non-linear tax incentives such as the Earned Income Tax Credit (Chetty and Saez 2013) or Social Security benefits (Liebman and Luttmer 2015) impact labor supply. Related work examines the impacts of providing information on fuel efficiency (Allcott and Knittel 2019), energy costs and savings (Kahn and Wolak 2013; Allcott and Taubinsky 2015; Davis and Metcalf 2016; Allcott and Sweeney 2017), the benefits of school choice or post-secondary education (Hastings and Weinstein 2008; Jensen 2010; Barr and Turner 2018), the returns to job search or retirement savings plans (Saez 2009; Duflo et al. 2016), restaurant hygiene grades (Jin and Leslie 2003), and truth-in-lending styles disclosures (Seira et al. 2017). Amongst this body of work, our study is most closely related to studies by Bollinger et al. (2011), Deb and Vargas (2016), and Restrepo (2017) exploring the impact of calorie labels on calories purchased or BMI. Our paper extends this earlier work by developing a theoretical model and empirical strategy to identify the channels through which calorie labels impact choice and by refining the construction of treatment and control groups.

Second, our paper contributes to a growing body of literature using subjective well-being to estimate the effects of government policies and economic shocks. One strand of this literature has used subjective well-being to place monetary value on public goods (or public bads) such as airport noise (Van Praag and Baarsma 2005), flood disasters (Luechinger and Raschky 2009), traffic congestion (Anderson et al. 2016), and air pollution (Luechinger 2009; Levinson 2012). Another strand has used subjective well-being to evaluate the impacts of inflation targeting (Di Tella et al. 2001), German reunification (Frijters et al. 2004), creative destruction (Aghion et al. 2016), cigarette taxes (Gruber and Mullainathan 2005), and improved housing for the poor

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(Galiani et al. 2018). Recent work shows that self-reported happiness measures are good predictors of future behavior (Kaiser and Oswald 2022) and provides a conceptual framework that maps shocks (or new information) into changes in stated happiness (Kimball and Willis 2023). Our paper extends this line of inquiry by using subjective well-being to test theory and identify channels through which a policy impacts choice rather than a means to recover preferences or estimates of welfare.

Third, our paper contributes to the literature on behavioral welfare economics. Although this literature has emphasized the importance of heterogeneity in the population (DellaVigna et al. 2012; Allcott and Taubinsky 2015, 2017; Taubinsky and Rees-Jones 2018; Allcott and Kessler 2019), it has largely ignored potential heterogeneity in the effects of nudges on consumer welfare. An exception to this is Allcott et al. (2022), who consider and test for heterogeneous responses to nudges and the resulting impact on welfare.[2] By exploring such heterogeneity, our paper is similar to Butera et al. (2022), who find that social recognition triggers different reactions for high (pride)- and low (shame)-performing individuals and Thunström (2019), who documents heterogeneity in stated emotional reaction to information on calorie labels. Our study differs from Thunström (2019) in that we examine behavior in a field setting and find that calorie labels negatively impact those with low internalities (proxied by BMI), suggesting that the nudge triggers moral costs as opposed to positive emotions.

Finally, our research contributes to a growing literature exploring how best to use insights from behavioral economics to achieve policy goals. To date, such work has focused on areas such as female labor force participation rates (Bursztyn et al. 2020), electricity and water conservation (Allcott 2011; Costa and Kahn 2013; Ferraro and Price 2013; Allcott and Rogers 2014), tax compliance (Fellner et al. 2013; Dwenger et al. 2016; Hallsworth et al. 2017), traffic violations (Chen et al. 2017), and the use of credit cards (Seira et al. 2017).[3] Unlike this prior work, our paper goes beyond simply evaluating whether or not the given policy works. We set forth to understand why the policy works and if this reason differs across segments of the population. In this regard, our paper shares similarity with work on the design of "sin" taxes (Gruber and Koszegi 2004; Taubinsky and Rees-Jones 2018; Allcott et al. 2019) and efforts to "nudge" the repayment of credit card debt (Bursztyn et al. 2019).

2. Background on Calorie Labeling Laws

Table 1 displays information on the passage and implementation of local calorie labeling mandates throughout the USA. New York City was the first location in the USA to pass a chain restaurant calorie labeling law. The New York City law was implemented and enforced starting in April 2008. Other areas across the country—King County, WA; Philadelphia County, PA; Albany and Schenectady County, NY;

2. Allcott et al. (2022) focus on the effect of nudges on the variance of the distortions as a source of welfare loss, whereas we focus on the welfare loss induced by an increase in moral costs.

3. For an overview of work in this area, see survey articles such as Downs et al. (2009) or Chetty (2015).

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TABLE 1. Menu labeling laws in localities prior to the implementation of the National Law in 2018.

Localities	Date passed	Date implemented	Requirement on number of establishments
New York City (New York, Bronx, Queens, Kings, Richmond Counties, NY)	December 2006	April 2008	≥15
Seattle (King County, WA)	August 2007	January 2009	≥15
Westchester County, NY (NYC MSA)	November 2008	June 2009	≥15
Philadelphia City/County, PA	December 2008	January 2010	≥15
Albany County, NY	September 2009	April 2010	≥15
Montgomery County, MD (DC MSA)	December 2009	July 2010	≥20
Schenectady County, NY (Albany MSA)	March 2010	October 2010	≥15
Suffolk County, NY (NYC MSA)	November 2009	November 2010	≥15
State of Vermont	June 2010	January 2011	≥20

Montgomery County, MD; and the state of Vermont—implemented similar laws between 2009 and 2011. Several other states and localities passed laws that were either repealed or never enforced due to anticipated preemption by the nationwide calorie labeling mandate within the affordable care act (ACA).[4]

We worked with a health policy expert to distinguish laws that were implemented, enforced, and never repealed from those that were either repealed or never enforced. The expert supplemented an exhaustive review of legal statutes and media articles with direct correspondence with state and local health departments. This process revealed substantial variation in the level and duration of enforcement of the implemented laws. While many locations actively enforced the local mandate throughout our sample period, enforcement in King County, Vermont, and Albany County was either weak or varied.

The initial studies exploring the impacts of these laws utilized receipt data and difference-in-difference designs.[5] Treatment groups were comprised of customers at restaurants exposed to a law and control groups constructed using customers eating at the same restaurant chain in neighboring areas. Results from this work are mixed. Elbel et al. (2009, 2011) find no evidence that the New York City law influenced calories ordered in low-income neighborhoods. Bollinger et al. (2011), in contrast, find that the New York City law reduced calories ordered at Starbucks restaurants by 6%. The evidence from King County, WA is also mixed—Finkelstein et al. (2011) report no effect on calories ordered at Taco Time Northwest restaurants whereas Krieger et al.

4. The 2010 ACA introduced a national mandate requiring chain restaurants and retail food establishments with 20 or more locations to post calories on menus and menu boards. The start date of this mandate was repeatedly delayed but ultimately implemented in 2018. Policymakers in many jurisdictions where mandates had passed shortly before the ACA never implemented them because of uncertainty about the final rules of the ACA mandate and the potential costs to restaurants to adopt certain menu labels only to change them when the ACA mandate took effect.

5. A related experimental literature suggests that providing nutritional information can affect meal decisions (e.g. Wisdom et al. 2010; Bedard and Kuhn 2015; VanEpps et al. 2016; Cawley et al. 2020).

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(2013) estimate a reduction in calories ordered at taco and coffee restaurants but not burger and sandwich restaurants.[6]

The mixed results suggest limitations inherent with analyses focusing on a small number of restaurants in a particular geographic area. Consumer preferences and knowledge may vary across locations, and the accuracy of beliefs about calories may differ across different types of restaurants. The ambiguity could also reflect differential and unaccounted for patterns of sorting whereby calorie labeling influences the types of people who eat at particular types of restaurants. Inconsistent enforcement could lead to unobserved heterogeneity in the likelihood any given establishment posted calories. Finally, such analyses are susceptible to concerns surrounding common trends and understated standard errors in comparative case studies with a small number of treated units (Abadie et al. 2010).

More recent studies shift the focus from estimating impacts on calories ordered to BMI. Restrepo (2017) compared changes in BMI in the counties in New York State that implemented calorie labeling laws to changes in other counties in the state and shows that local laws reduced average BMI by 0.4 kg/m^2 (or nearly 1.5%) and the probability of being obese by 3 percentage points. Yelowitz (2016) used respondents from the thirty largest US cities, including several with calorie mandates, and documents small and statistically insignificant reductions in both average BMI and the probability of being obese. Deb and Vargas (2016) utilized a data-driven entropy balancing procedure to reweight the control group to best match the treatment group's pre-treatment BMI trajectory and finds small reductions in BMI for both men and women.

Our approach differs from this prior work along multiple dimensions. First, we develop a more rigorous classification of treatment and control groups. Prior work identified treatment groups utilizing only information on the passage and intended effective dates of local calorie labeling laws. In doing so, we conclude that many of the "treated" areas of Yelowitz (2016) and Deb and Vargas (2016) were either never actually treated (e.g. Oregon) or only temporarily treated (e.g. California). We define treated areas as those where the mandate was implemented as intended and never repealed, and subdivide treated areas based on level of enforcement. We validate these choices by making phone calls to restaurants and conducting consumer surveys to document clear differences in the prevalence of calorie labeling between treatment sites and neighboring areas.

A second difference between our approach and the prior literature is that we evaluate calorie-labeling laws through the lens of a theoretical framework whereby changes in consumer behavior need not lead to increased consumer welfare. This differs from the standard economic model whereby calorie labeling laws only influence choices by rectifying imperfect information.[7] In contrast, our framework allows for calorie labels to trigger psychological or moral costs that are akin to a tax on high

6. Auchincloss et al. (2013) is the only paper to estimate the effect of the mandate in Philadelphia and finds that customers eating in treated restaurants consume 151 fewer calories than do counterparts eating at the same chain in surrounding areas.

7. Such sentiment is highlighted in Bollinger et al. (2011) who write that, "…as far as regulatory policies go, the costs of calorie posting are very low—so even these small benefits could outweigh the costs."

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calorie meals. The net effect of labeling laws on consumer welfare is thus theoretically ambiguous *but* can be inferred using available data. By evaluating this possibility, we extend prior work and look not only at changes in BMI but also changes in subjective well-being and stated feelings of guilt and social appropriateness.

3. Theoretical Model

It is well known that the environment (or *choice architecture*) in which people make decisions can affect their choices (Thaler and Sunstein 2008). We assume individuals make two choices: a meal $x \in X$ given a menu X, and calories consumed outside the restaurant $\bar{\kappa} \geq 0$. Choices are made in a given environment $e \in \{0, 1\}$, where $e = 0$ represents the baseline environment and $e = 1$ represents the introduction of calorie labels. We represent the calories of meal x by $k(x)$ and total calories consumed as $k(x) + \bar{\kappa}$.

We build on the framework developed in Jimenez-Gomez (2023) to model the calorie label nudge. We consider a representative individual i, but we omit the subindex i in what follows for notational clarity. We define "experienced utility" $u^{exp}(x, \bar{\kappa}|e)$ or the "hedonic experience associated with an outcome" (Kahneman and Thaler 2006) as a function of calories consumed and the health effects associated with changes in weight due to calories consumed. People, however, do not always maximize their experienced utility (Kahneman et al. 1997; Berridge 2003). Formally, we define "decision utility" $u^{dec}(x, \bar{\kappa}|e)$ as the utility that individual i maximizes when selecting a meal x and calories $\bar{\kappa}$ (Allcott and Taubinsky 2015).

Critics of nudges have argued that they might impose moral costs (Glaeser 2006). As described by Levitt and List (2007), utility may reflect both consumption and psychological or moral costs for violating social norms. Nudges can increase such moral costs, an impact that has been documented in a variety of experimental contexts. For instance, Andreoni and Rao (2011) and DellaVigna et al. (2012) document reluctant giving whereby individuals donate to alleviate a "shadow tax" of guilt that arises when asked to give. Conditional on being solicited, individuals may find it optimal to give, but they wind up worse off than if they had never been solicited. A parallel process could arise through the introduction of calorie labels on menus and menu boards. Observing that a preferred menu item has a high calorie count could trigger a "shadow tax" or guilt that can only be alleviated by ordering a lower calorie alternative. However, doing so leads to lower levels of utility than would be realized had the individual not seen the calorie label and ordered the more preferred, higher calorie item.

We incorporate such a moral cost by assuming that experienced utility u^{exp}, the concept relevant for welfare analysis (Kahneman and Sugdgen 2005; Allcott and Taubinsky 2015), is

$$u^{exp}(x, \bar{\kappa}|e) = v(x) + w(k(x) + \bar{\kappa}) - c^e(b^e(x)),$$

where $v(x)$ is the utility from characteristics of the meal such as taste and nutritional value; $w(k(x) + \bar{\kappa})$ represents utility derived from total calories consumed; and

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$c^e(b^e(x))$ is the moral cost from consuming highly caloric foods. We assume moral costs are increasing and convex and depend on the environment—for example, whether calories are posted—both directly and through beliefs $b^e(x)$ about the calorie content of meal x.[8] Following Philipson and Posner (1999), we assume $w(\kappa)$ is strictly concave and has a maximum at $\kappa = \kappa^*$ such that utility is increasing in weight up to an inflection point and decreasing afterward.

We model the individual's decision utility as

$$u^{dec}(x, \bar{\kappa}|e) = v(x) + w(b^e(x) + \bar{\kappa}) - c^e(b^e(x)) + \Lambda^e(k(x)).$$

Decision utility differs from experienced utility in two aspects. First, the function $w(b^e(x) + \bar{\kappa})$ depends on the beliefs about calories from meal x, not on the actual caloric content $k(x)$. Second, there is a non-informational internality $\Lambda^e(k(x))$ that depends on both the actual calories and the environment. The non-informational internality is an increasing function of calories that encompasses several sources of non-utility-maximizing behavior such as lack of attention, limited "cognitive bandwidth," self-control problems, or time inconsistency (Loewenstein et al. 2003; Mullainathan and Shafir 2013; Courtemanche et al. 2015a; Bradford et al. 2017; Allcott et al. 2019; Sadoff et al. 2020). We assume that, given an environment, each individual has a unique preferred combination of meal and calories consumed outside of the restaurant.[9]

Calorie labels could influence a decision-maker's choice through three channels. First, calorie labels may provide information that influences choice through $w(b^e(x) + \bar{\kappa})$. Second, calorie labels could also reduce non-informational internalities $\Lambda^e(k(x))$. This could occur by increasing salience of calories (reducing the extent to which they are shrouded) relative to having to consciously look on a restaurant's website, thus alleviating bias from inattention or cognitive bandwidth. Additionally, calorie labels could serve to slow down one's thinking and, hence, the extent to which choices reflect "emotional" responses (e.g. self-control problems; time-inconsistency). Finally, calorie labels can increase moral costs $c^e(b^e(x))$.

We obtain predictions that can be tested with the BRFSS by imposing these assumptions:

1. Individuals have at least as large a marginal non-informational internality in the status quo environment without the nudge, $\Lambda^0(y) - \Lambda^0(y') \geq \Lambda^1(y) - \Lambda^1(y')$ for all $y > y'$.[10]

8. In Online Appendix A, we consider a general model that includes prices for the meal and for calories consumed outside of the restaurant. Results in this section (Propositions 1 and 2) hold in this more general setup.

9. This assumption is used for technical convenience. It could be dispensed with if either Assumption 1 or 2 below was stated as a strong rather than a weak inequality. We discuss this further in Online Appendix A.

10. Note that this assumption allows for "over-sensitization", as in Allcott et al. (2022), in the sense that some consumers could go from a positive (in $e = 0$) to a negative (in $e = 1$) marginal non-informational

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3. The environment with the nudge has a (weakly) larger marginal moral cost, that is, $c^0(y) - c^0(y') \leq c^1(y) - c^1(y')$ for all $y > y'$.
4. Beliefs have the following structure: $b^e(x) = a^e k(x)$, with $a^0, a^1 \geq 0$ and $a^1 = \gamma + (1-\gamma)a^0$, for $\gamma \in [0, 1]$. This structure implies that the nudge—calorie labels—makes beliefs (weakly) more accurate.
5. At baseline (in $e = 0$) individuals do not underconsume (i.e. they consume at least the optimal amount κ^* of calories), and they do not overestimate calories, that is, $b^0(x) \leq k(x)$ for all x.

These assumptions place minimal structure on observed choice but allow for the derivation of concrete predictions regarding changes in calories consumed and the welfare effects of calorie mandates.[11] We now explore the effect of calorie labels on the decisionmaker's choice and the resulting welfare implications. Proofs of the results can be found in Online Appendix A.

PROPOSITION 1. *Calorie labels (weakly) reduce the amount of calories consumed at the restaurant, as well as the total amount of calories consumed.*

The intuition for Proposition 1 is that, with calorie labels, the individual (at least weakly) (i) improves the accuracy of their beliefs, (ii) has their marginal non-informational internality reduced, and (iii) has their marginal moral cost increased. The fourth assumption ensures that these effects go in the same direction. The presence of calorie labels makes the individual less susceptible to biases favoring the selection of higher calorie meals which also impose a higher moral cost.[12] Thus, the individual consumes fewer calories at restaurants that post calories, and the same effects discussed above ensure that the individual increases calories outside of the restaurant by at most that amount, so that total calories consumed also fall.

We have described the model using a representative decisionmaker. However, heterogeneity in the population has been shown crucial for behavioral welfare economics (Allcott and Taubinsky 2015; Taubinsky and Rees-Jones 2018; Allcott and Kessler 2019). For simplicity, suppose that the population can be divided in two groups, A and B. Individual $a \in A$ is an agent for whom informational and non-informational internalities have little to no discernible impact on choices in the baseline environment, for example, $\Lambda_a^0(x) = 0$ for all $x \in X$ and $b_a^0(x) = k(x)$. On

internality. For such consumers, the nudge would lead to an over-correction in the decrease of calories consumed.

11. Note that while Proposition 1 requires all four assumptions, Proposition 2 only requires Assumption 2.

12. We would like to thank a referee who correctly pointed out that overconsumption plays an important role in this result. In the absence of overconsumption (i.e. if the individual was choosing meals that are less caloric than what is optimal), it is possible that calorie labels would actually *increase* calorie consumption, if the individual attempts to maximize the amount of calories per dollar spent (and had ex-ante inaccurate beliefs).

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the other hand, an individual $b \in B$ is an agent for whom informational and non-informational internalities can have a discernible impact on calories consumed in the baseline environment.

The welfare gain/loss from the nudge for individual i is the difference in experienced utility from choices made with calorie labels ($e = 1$) versus choices in the baseline environment ($e = 0$). Intuitively, we would expect the nudge to lead to a reduction in welfare of an individual $a \in A$. For such agents, the potential benefits of the nudge—reduced internalities—are negligible and likely dominated by increased moral costs. In contrast, the welfare effects for an individual $b \in B$ are ambiguous; the net effect depends on the relative welfare gains from reduced internalities and the welfare loss from increased moral costs. This intuition is formalized in the following proposition:

PROPOSITION 2. *The following predictions hold true for the introduction of calorie labels:*

1. *Those in A will have a welfare loss (or no change) from the nudge .*
2. *The welfare effect in B is ambiguous and depends on the relative size of the gains from information, the reduction in the non-informational internality, and the moral cost associated with the nudge.*

An immediate implication of Proposition 2 is that the average welfare effect in a population is ambiguous. However, we would expect the overall impact to be positive when either (i) the impact of internalities and misinformation on choices and overconsumption by those in the B group is substantial or (ii) the proportion of B-types in the population is large. In contrast, we would expect the nudge to lead to decreased social welfare when either (i) the change in moral costs (for either group) is large or (ii) there is a high proportion of A-types in the population.

In summary, our theoretical framework generates the following predictions. The introduction of calorie labels will lead (at least weakly) to (i) a decrease in average calories consumed, (ii) a decrease in welfare for a segment of the population (type A consumers), and (iii) an ambiguous change in welfare for another segment of the population (type B consumers). Further, our theoretical framework yields predictions to test the relative importance of the channels through which calorie labels impact consumer choice and well-being. If calorie labels work predominantly through the information and/or non-informational internality channels, the average effect on life satisfaction would be unambiguously positive. The introduction of such labels would have no impact on the choices and welfare of Type As and reduce internalities that drive overconsumption by Type Bs. Alternately, if the main impact of calorie labels is increased moral costs, the effect for Type As is unambiguously negative while the effect for Type Bs is ambiguous.

To test the model predictions and uncover the relative importance of the various channels, we combine primary and secondary data to examine the impact of calorie

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labels on several outcomes of interest: (i) BMI, which is a proxy for calories consumed, (ii) self-reported measures of life satisfaction, which is a proxy for utility or consumer well-being, and (iii) measures of psychological (or moral) costs elicited via a survey experiment. We describe the data in Section 4; the analysis of the average effect on BMI in Section 5; and the analysis of the relative importance of the channels examining heterogeneity in the impact on BMI, life satisfaction, and measures of moral costs in Section 6.

4. Data

4.1. Identification of Treatment and Control Areas

Our identification strategy compares counties that implemented calorie labeling laws to proximate areas that did not implement such laws. Our main analysis sample includes individuals residing in the Metropolitan Statistical Areas (MSAs) of New York City (the "treated" counties are the five counties of New York City, Suffolk County, and Westchester County; the control counties are the other counties in the area), Philadelphia (treated county is Philadelphia), Seattle (treated county is King), Washington DC (treated county is Montgomery), and Albany (treated counties are Albany and Schenectady) as well as the entire census division of New England (treated state is Vermont).[13] We exclude from the sample observations from individuals residing in Maine, Massachusetts, New Jersey, and Nassau County in New York, where menu labeling laws were passed but never enforced, due to possible anticipatory responses.

Using proximate geographic areas as control groups increases the likelihood that the treatment and control groups experience similar local labor market and economic conditions—especially those that could impact BMI and stated well-being independent of exposure to calorie labels. This is particularly important as the implementation of many of the local calorie mandates occurred shortly after the Great Recession. Similarly, individuals in proximate geographic areas are in the same media markets and experience similar public health messaging.

One concern with our identification strategy is that restaurant chains may voluntarily post calories on menus in establishments in our control areas. Thus, individuals in our control areas may be exposed to calorie labels even when eating at select local chains. Further, customers who reside in the control areas may dine at restaurants in treated regions and, thus, be exposed to calorie counts. If these concerns are valid, they should serve to minimize differences across respondents from treated and control areas and thus attenuate the estimated effects. Therefore, our estimates

13. Ulster County, NY also passed and enforced a menu labeling law, but we exclude it since it is the only county within its MSA. The results are robust to including it, either with or without adjacent Orange County as a control.

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should be considered intent-to-treat effects and a lower bound on the impacts of calorie labels.

To assess the first concern, we called 15,099 restaurants in our sample areas to inquire whether they were currently posting calories on menus. These restaurants encompassed all locations of the top 500 chains by sales volume.[14] The calls were made in spring 2017, prior to the implementation of the federal calorie mandate. Of the 2,305 restaurants in the regions that were actively enforcing local mandates, 83% were compliant with the local mandate and posting calories on menus and menu boards. Of the 1,199 restaurants in the regions that had mandates but were not actively enforcing them, 73% were posting calories. Of the 11,595 restaurants in the control regions, 58% were voluntarily posting calories. Therefore, as of spring 2017, there were clear differences in the prevalence of calorie posting between treated and control regions and amongst treated regions with active enforcement versus those with inactive enforcement. Moreover, the observed differences likely understate the "true" differences present during our sample period, which ends in 2012. By 2017, many chain restaurants voluntarily posted calories on menus in all locations. In contrast, McDonald's was the only chain to do so nationwide during our sample period, and this did not occur until September 2012.[15]

To examine the prevalence with which consumers observe calories on menus, we surveyed 5,228 adults throughout our sample region in December 2014. The survey, denoted as Survey 1, was implemented in collaboration with Qualtrics Panels to construct a representative sample based on age, gender, and education. Respondents were asked whether they had noticed calories on the menu or menu board of each restaurant they had eaten at during the previous 24 hours. Amongst respondents living in regions actively enforcing local calorie-posting mandates, 48.9% reported that they had noticed calorie counts on the menu. In regions not actively enforcing local mandates, 41.1% of respondents noticed calories on menus, and in regions without a local mandate, 36.0% of respondents reported noticing calories. Hence, as of 2014, individuals residing in areas actively enforcing calorie-posting requirements were 36% more likely to have observed calorie counts when eating at a restaurant than were counterparts residing in control areas. Again, the observed difference is likely smaller than the difference during our sample period.[16]

14. The top 500 restaurant chains by sales volume are drawn from the 2016 Technomic Top 500 Chain Restaurant Advance Report. Given that many of the chains outside of the top 400 by sales volume have less than 15 locations, these 500 chains likely encompass nearly all restaurants required to post calories by the local mandates.

15. See, for instance, https://www.npr.org/2012/09/13/161050157/mcdonalds-to-post-calories-on-menu-boards. By 2017, at least Au Bon Pain, Bahama Breeze, Chick-fil-A, Dunkin Donuts, Firehouse Subs, Jamba Juice, Panera Bread, Starbucks, and Subway also posted calories on menus and menu boards in all locations nationwide.

16. Data from another wave of the survey conducted in December 2016 reinforce that the Wave 1 results likely understate the extent of heterogeneity in our sample period. We observe a larger increase in the percentage of respondents reporting that they noticed calories in control areas than in areas with pre-existing local mandates.

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Ideally, we could use data from the phone calls and Survey 1 as first-stage estimates to recover a local average treatment effect for the effects of calorie-posting requirements on calories consumed in chain restaurants. However, the mismatch of timing between the phone calls and surveys and our BRFSS sample period prevents such an analysis. Rather than interpreting Survey 1 data as first-stage estimates, they are better viewed as evidence that: (i) restaurants comply with local mandates, (ii) compliance depends on the stringency with which local mandates are enforced, (iii) customers notice the requirements, and (iv) the likelihood an individual notices calorie labels depends on the stringency of enforcement in the area where they reside.

4.2. BRFSS Data, Outcome Measures, and Controls

Our primary data source is the BRFSS for the period between 1994 and 2012. The BRFSS is a random-digit-dialing telephone survey conducted by state health departments and the Centers for Disease Control and Prevention (CDC 1994–2012). The BRFSS collects information about adults' health, health-related behaviors, and demographic characteristics. Importantly, the BRFSS includes state and county identifiers, although the latter are not publicly available after 2012. However, we view ending the sample in 2012 as an *advantage* for our research question given the aforementioned voluntary nationwide posting by many chains after 2012.

We construct our main analysis sample as follows. Pooling all observations from 1994 to 2012 yields 5,232,786 respondents. Restricting geography to the treatment and control areas leaves 614,179 respondents. From this sample, we exclude 5,608 pregnant women and 14,207 individuals with missing information. The final sample consists of 594,364 individuals. Online Appendix Table B.1 displays the composition of the sample by county.

Our theoretical model provides testable predictions for calorie consumption and welfare. We do not observe these outcomes directly and instead use BMI and life satisfaction as proxies for them.[17] BMI is computed based on self-reported measures as weight in kilograms divided by the square of height in meters.[18] Provided that calorie expenditure is unaffected by treatment, the predicted effects of calorie labeling on caloric intake and BMI are of the same sign (negative).

17. The only government survey dataset with a comprehensive food diary—the National Health and Nutrition Examination Survey (NHANES)—is not nearly large enough and does not have the geographic scope to precisely identify the impact of the local calorie mandates of interest in our study.

18. Self-reported weight and height admittedly suffer from measurement error. Cawley (2004) and Courtemanche et al. (2015b) develop "corrections" that adjust self-reports based on the comparison of actual and self-reported values in the NHANES. We elect not to implement these corrections. Our sample period does not correspond in a straightforward way to the NHANES survey periods and assumptions to reconcile the difference could potentially induce more error than the correction solves. Further, research into the causes of obesity has shown that implementing such corrections has no substantive impact on results (Courtemanche et al. 2015b, 2016). Finally, the nature of reporting error is unlikely to differ across treatment and control areas.

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Given that the BRFSS data do not include food consumption or calories consumed, we use information on other health-related behaviors—smoking, alcohol consumption, and exercise—to determine whether any observed impacts on BMI reflect changes in our unobserved measure of interest (calories consumed) or a broader array of behavioral change that would influence net caloric intake. Smoking is measured as whether the individual currently smokes cigarettes. Alcohol consumption is the number of drinks consumed during the past 30 days. The physical activity outcomes are any exercise in the past 30 days besides a regular job, the average minutes of moderate exercise per week over the past month, and the average minutes of vigorous exercise per week.[19] The latter two outcomes were only collected in odd years between 2001 and 2011.

The BRFSS includes a question about life satisfaction, with possible responses being very satisfied, satisfied, dissatisfied, and very dissatisfied. We assign these responses values of 3, 2, 1, and 0, respectively. Following Stevenson and Wolfers (2013) and Killingsworth (2021), we standardize life satisfaction so that the coefficient estimates can be interpreted in terms of standard deviation changes. The life satisfaction variable is only available from 2005 to 2010, so we exclude the Washington, DC MSA and the census division of New England as there is no data on life satisfaction in the post-treatment period in those areas.[20]

Life satisfaction has been used in prior work to evaluate the effects on consumer well-being of policy changes. While the use of self-reported life satisfaction is not beyond reproach (Deaton and Stone 2016), it has been proposed as a useful proxy for changes in experienced utility (Kahneman and Sugden 2005), and evidence suggests that it is reliable for this purpose.[21] Stated measures of subjective well-being are a powerful predictor of stated choices (Benjamin et al. 2012) and negatively correlated with more objective measures such as demand for antidepressants (Blanchflower and Oswald 2016; Kaiser and Oswald 2022). Kaiser and Oswald (2022) use longitudinal

19. Restrepo (2017) measures physical activity as any exercise in the past 30 days and the number of minutes of any form of exercise in the past 30 days; he finds no impact of the New York City law on physical activity.

20. The treated county in the Washington DC MSA—Montgomery County, MD—implemented its calorie mandate in July 2010. However, all respondents from Montgomery County in the 2010 wave of the BRFSS were surveyed before then. The treated state in New England—Vermont—did no implement its mandate until January 2011.

21. Kimball and Willis (2023) develop a conceptual framework that provides support for such an interpretation. They model life-satisfaction (or stated happiness) as the sum of two components: (i) elation or a transitory response to recent news/shocks that impact lifetime utility and (ii) baseline mood or long-run happiness. The introduction of calorie mandates is a shock that could impact lifetime utility and hence stated life-satisfaction. The direction of this change reveals whether the average subject perceives that the policy will lead to increases or decreases in lifetime utility and provides information as to channels through which the labels impact choice.

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data to show that lower life satisfaction in one period increases "get-out-of-here" behaviors such as changing jobs or moving in the next period.[22]

The BRFSS data also include demographic and socioeconomics characteristics of the respondents, for example, race/ethnicity, educational attainment, age, gender, and marital status. As these variables are likely correlated with our outcomes of interest, we use them as controls in our empirical models. The race/ethnicity variables are indicators for whether the individual is non-Hispanic White (the reference category), non-Hispanic Black, Hispanic, or another race/ethnicity. Educational attainment is measured by four mutually exclusive categories for whether the individual did not obtain a high-school degree (the reference category), obtained a high-school degree, attended college but did not graduate, or obtained a college degree. We include dummy variables for each individual year of age along with indicators for females and married individuals. In addition, we control for county-level socioeconomic characteristics that may correlate with our outcomes of interest. We measure annual population density as the number of people per square mile using U.S. Census Bureau information on annual population estimates and land area estimates from 2000 (U.S. Census Bureau, 2014 a, b). Annual median household income estimates come from the Census Bureau (U.S. Census Bureau, 2014 c) and monthly unemployment rates from the Bureau of Labor Statistics (U.S. Bureau of Labor Statistics, 2014).

5. Effects on Body Mass Index

Our theoretical model predicts that *all* consumers will reduce calories consumed when exposed to calorie counts on menus. We evaluate this prediction by estimating the average effect of chain restaurant calorie posting mandates on our proxy measure of calories consumed, BMI. In Section 5.1, we describe our estimating equation and identification strategy. Section 5.2 presents evidence on the validity of the comparison group. Section 5.3 presents the estimated average effects on BMI, while Section 5.4 provides evidence to support a causal interpretation of those estimates. Section 5.5 investigates other outcomes that could influence BMI in the absence of changes in calories consumed.

5.1. Econometric Model

We estimate the following empirical model:

$$Y_{ict} = \beta_0 + \beta_1 LAW_{ct} + \boldsymbol{\beta_2 X_{ict}} + \boldsymbol{\beta_3 Z_{ct}} + \alpha_c + \tau_t + \gamma_c t + \varepsilon_{ict}, \tag{1}$$

where Y_{ict} is the outcome variable (BMI) for individual i in county c in time period t. Since survey data are collected monthly and we have the exact month and year when the

22. A major concern with the use of such questions is that observed responses are often sensitive to irrelevant cues that could confound the interpretation of differences in outcomes across subsets of data. Heffetz and Reeves (2019) show that inference drawn from life satisfaction questions in the BRFSS data does not appear to be influenced by the way in which the data were collected.

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laws were implemented, the time unit is "month-year" (e.g. April 2008). The treatment indicator, defined as LAW_{ct}, is equal to one whenever a chain restaurant calorie labeling law is implemented in county c in month-year t. The associated coefficient β_1 measures the average treatment effect on the outcome. $\boldsymbol{X_{ict}}$ denotes the individual-level controls listed above, and $\boldsymbol{Z_{ct}}$ denotes the county-level controls. Equation (3) also includes county fixed effects α_c, month-year fixed effects τ_t, and county-specific linear time trends $\gamma_c t$. The last term, ε_{ict}, is an idiosyncratic error term. Standard errors are clustered at the county level and are adjusted for heteroscedasticity. As there are 110 counties in our sample, concerns about inference with a small number of clusters are reduced (Cameron and Miller 2015). We utilize sampling weights in all analyses to control for changes in sample composition resulting from the incorporation of cell phones with the 2011 wave.

Our identification strategy accounts for an array of observable characteristics as well as unobservable characteristics that are constant within counties and time periods or change linearly within counties. In other words, $\hat{\beta}_1$ gives the causal effect of menu labeling laws under the assumption that the implementation of such laws is uncorrelated with unobservable county characteristics that change non-linearly. We check the plausibility of this identification strategy by examining whether individual characteristics in the treated and control regions are similar prior to the implementation of the first calorie-posting requirement and whether the trends in BMI are parallel for these two groups during the pre-treatment period. We also conduct an event study analysis and a series of robustness and falsification tests.

5.2. Comparison of Treatment and Control Groups

Table 2 presents summary statistics for the outcome variables and covariates in the pre-treatment period (1994–2007) for the treated and control groups. Weight outcomes in the treated and control groups were similar prior to the implementation of calorie labeling laws: average BMI was around 26.1 kg/m^2 while 36% were overweight but not obese ($25 \leq$ BMI < 30), 12%–13% were Class I obese ($30 \leq$ BMI < 35), and 6% were Class II/III obese (BMI ≥ 35). Pre-treatment life satisfaction was somewhat lower in the treatment group than the control group with standardized values of -0.101 and 0.055, respectively.[23] Whereas education, age, and gender are similar across treatment and control groups, the control group had a smaller proportion of racial/ethnic minorities and lived in higher income counties that were less densely populated.

Given the large sample sizes, nearly all differences in the means are statistically significant. Table 2 also reports standardized differences, which are differences in the means by treatment status divided by the square root of the sum of the variances;

23. Using the synthetic control method, the baseline difference between life satisfaction across groups is reversed. The control group has slightly lower life satisfaction, with a mean of –0.162 compared to a mean of –0.112 in the treatment group. Nonetheless, the synthetic control method results are similar to those from the baseline model.

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TABLE 2. Pre-treatment means with standard deviations in parentheses.

Variable	Treatment group	Control group	Standardized difference
Body mass index (BMI = weight in kg/height in m^2)	26.093	26.177	−0.012
	(5.195)	(5.098)	
BMI categories			
Healthy weight (BMI < 25)	0.466	0.453	0.018
	(0.499)	(0.498)	
Overweight ($25 \leq$ BMI < 30 kg/m^2)	0.356	0.362	−0.009
	(0.479)	(0.481)	
Class I obese ($30 \leq$ BMI < 35 kg/m^2)	0.122	0.128	−0.011
	(0.328)	(0.334)	
Class II/III obese (BMI ≥ 35 kg/m^2)	0.055	0.057	−0.004
	(0.229)	(0.231)	
Other health behaviors			
Currently smokes cigarettes	0.201	0.208	−0.013
	(0.401)	(0.406)	
Number of drinks (0 if none) in past 30 days	9.496	11.534	−0.075
	(18.083)	(20.485)	
Any exercise (besides regular job) in past month	0.725	0.783	−0.096
	(0.447)	(0.412)	
Minutes per week over the past month of moderate exercise (large increases in heart rate/breathing)	48.793	52.497	−0.039
	(66.802)	(67.213)	
Minutes per week over the past month of vigorous exercise (small increases in heart rate/breathing)	30.859	35.113	−0.057
	(51.165)	(53.556)	
Life satisfaction measures			
Life satisfaction (standardized)	−0.101	0.055	−0.111
	(1.004)	(0.975)	
Very dissatisfied	0.013	0.010	0.020
	(0.113)	(0.099)	
Dissatisfied	0.051	0.036	0.051
	(0.220)	(0.187)	
Satisfied	0.537	0.478	0.084
	(0.499)	(0.499)	
Very satisfied	0.400	0.476	−0.110
	(0.490)	(0.499)	
Control variables			
Race/ethnicity is non-Hispanic Black	0.180	0.101	0.162
	(0.384)	(0.301)	
Ethnicity is Hispanic	0.166	0.054	0.258
	(0.372)	(0.226)	
Race/ethnicity is neither White, Black, nor Hispanic	0.092	0.049	0.120
	(0.289)	(0.215)	
High school degree but no further	0.268	0.282	−0.023
	(0.443)	(0.450)	
Some college but no 4-year degree	0.242	0.257	−0.024
	(0.428)	(0.437)	
College graduate or higher	0.363	0.381	−0.027
	(0.481)	(0.486)	
Age in years	44.883	45.421	−0.022
	(17.357)	(17.112)	
Gender is female	0.507	0.500	0.010
	(0.500)	(0.500)	

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TABLE 2. Continued

Variable	Treatment group	Control group	Standardized difference
Married	0.497	0.594	−0.140
	(0.500)	(0.491)	
County population density	19,537	1,520	0.819
	(21,897)	(1,988)	
County median household income	45,603	55,868	−0.515
	(15,335)	(12,749)	
County unemployment rate in percent	5.809	4.208	0.647
	(2.116)	(1.285)	
Sample size in pre-treatment period (1994–2007)	120,755	246,848	

Notes: Sample sizes vary for the other health behavior and life satisfaction outcomes. Observations are weighted using the BRFSS sampling weights.

a measure that does not vary with sample sizes (Imbens and Wooldridge 2009). The standardized difference exceeds the rule of thumb of 0.25 only for the percent Hispanic and county-level characteristics (population density, median income, and unemployment rate). Our identification strategy will control for these differences directly, while the county fixed effects and county-specific linear time trends will help account for any unobservable characteristics that exhibit similar differences. Yet, we extensively examine the robustness of the results to alternative specifications and sets of controls.

Figure 1 presents trends in average BMI in the treatment and control locations during our sample period. The vertical line, drawn between 2007 and 2008, reflects the timing of the first calorie labeling law, implemented by New York City in 2008. The treatment and control groups exhibited very similar levels and trends in average BMI in the pre-treatment period. This is consistent with the identifying assumption and suggests that the individuals in the control group provide a valid counterfactual for the outcomes of individuals in the treatment group. Following the implementation of the New York City law in 2008, there is a flattening in the trend of BMI change for individuals in the treated areas. This leads to differences in BMI for individuals in treated counties relative to counterparts in control areas. The relative difference increases in 2011 after the implementation of all remaining laws and ultimately peaks with a relative difference of approximately 0.5 BMI units by the end of the sample period.[24]

5.3. Average Effect of Calorie Labeling Laws on BMI

We next report estimates of the average effect of calorie labeling laws on BMI from the difference-in-differences design of equation (1) in column (1) of Table 3. We find

24. To put this into perspective, a difference of 2.5 pounds for women and 3 pounds for men would lead to a change in BMI from 27.0 to 27.5 if the individual was of average height.

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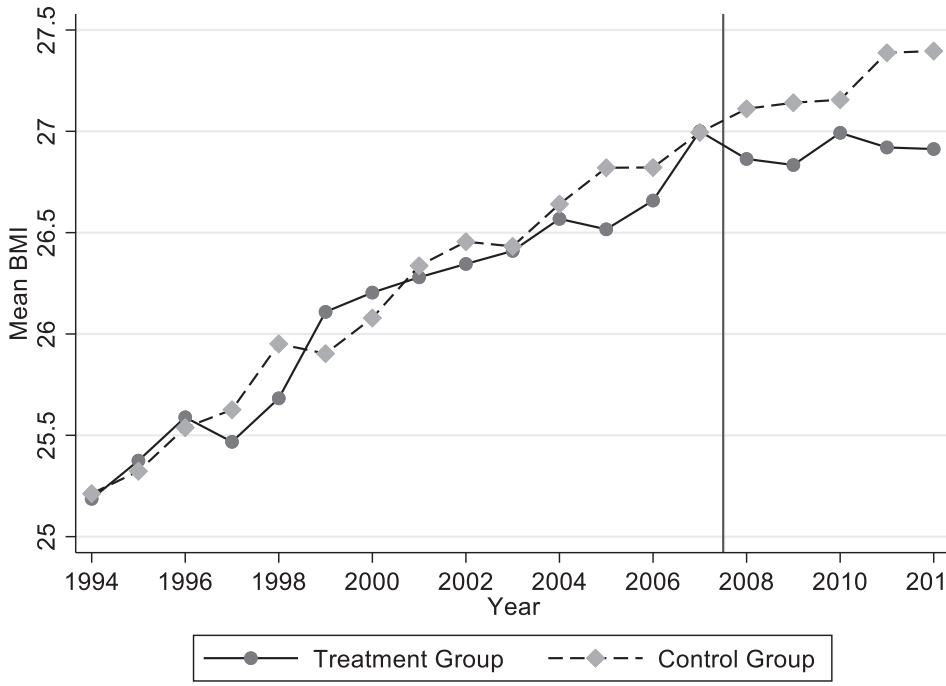

FIGURE 1. Average BMI during sample period. Observations are weighted using the BRFSS sampling weights and aggregated to the county level. The treatment group consists of all localities listed in Table 1, while the control group consists of other counties in MSAs with at least one "treated" county, plus New England, excluding any counties/states in these areas that passed but did not implement a calorie labeling law. The vertical line is drawn between 2007 and 2008, prior to the implementation of the first law in New York City in April 2008.

TABLE 3. Effects of chain restaurant calorie posting laws on body mass index.

	(1)	(2)
Law implemented	−0.174	−0.191
	(0.081)	(0.099)
Law implemented but no longer actively enforced		0.126
		(0.121)
Law passed but not yet implemented		0.001
		(0.099)
Sample size	594,364	594,364
Mean of outcome	26.711	26.711

Notes: Standard errors, heteroskedasticity-robust, and clustered by county, are in parentheses. Observations are weighted using the BRFSS sampling weights. The control group consists of other counties in MSAs with at least one "treated" county, plus New England, excluding any counties/states in these areas that passed but did not implement a calorie labeling law. The controls include binary indicators for the individual's sex, age, race/ethnicity, marital status, and education level, as well as county-level population density, median income, and unemployment rate. County and month∗year fixed effects are also included, as well as linear county-specific time trends.

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that calorie posting laws lead to a statistically significant (5% level) difference in BMI of 0.17 kg/m^2 for those in treated areas relative to counterparts in the control group. This relative difference in BMI corresponds to a 1.5-pound difference in weight at the mean height of the sample or a difference of 0.7% relative to the mean of the treatment group prior to the implementation of the laws.

Our estimated treatment effect corresponds to an approximate difference of 105 calories consumed per week by individuals residing in our treatment areas relative to counterparts residing in control areas.[25] In 2012, the average US household reported eating approximately 5 meals per week away from home. As 40% of restaurants in the USA were chain restaurants with at least 20 locations nationwide (Saskena et al. 2018), we would expect at least 2 meals per week would be consumed at a restaurant subject to a calorie mandate. Assuming individuals only change calories consumed when eating at chain restaurants, our estimated treatment effect corresponds to a difference of 53 calories per meal.[26]

Our estimated effects on BMI are lower than what would be predicted based on experimental estimates of the effect of calorie labels on calories consumed at restaurants (Wisdom et al. 2010; VanEpps et al. 2016; Cawley et al. 2020). For comparison, Wisdom et al. (2010) find that including calories and daily calorie recommendations on menus in a fast-food restaurant decreased calories ordered in a meal by 99 calories, VanEpps et al. (2016) find that including calories on online menus for a large corporate cafeteria led to a reduction of 60 calories per order, and Cawley et al. (2020) find that including calories on the menu in a full-service restaurant reduced calories ordered by 45 calories. Based on these results, we would expect a relative decrease in BMI of 0.75 kg/m^2, 0.47 kg/m^2, and 0.35 kg/m^2, respectively, within 1 year.

Viewed through the lens of work on scalability, such divergence is not surprising (Al-Ubaydli et al. 2017a, 2017b). By design of the local mandates, individuals can avoid calorie labels by choosing to eat at locations that are not required to post calories. If calorie labels impose a moral cost on the consumption of high calorie foods, we would expect some individuals to switch where they eat to avoid exposure. Such sorting would attenuate the effect by limiting the number of meals purchased at restaurants affected by the laws. Moreover, our data suggest imperfect compliance and hence reduced fidelity of the "treatment"; many chain restaurants in treated areas report that they do not post calories on menus.

Next, to better relate the average treatment effects reported in column (1) of Table 3 to the prevalence of calorie labels on menus (fidelity of treatment), we augment equation (1) to include variables denoting that the law was passed

25. This estimate is based upon Hall et al. (2011), who shows that a reduction of 10 calories consumed per day will eventually lead to a one-pound weight loss.

26. Our back of the envelope calculation likely understates the number of meals consumed per week at chain restaurants. According to our survey data described in Section 4.1 (Survey 1), individuals in our sample areas reported eating approximately 4 meals/snack a week at a chain restaurant—which would imply that our estimated treatment effect corresponds to a reduction of 26 calories consumed per meal.

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but not yet implemented and that the law was implemented but is not currently actively enforced. The first indicator assesses anticipatory effects from restaurants posting calories on menus before implementation of local mandates or public health messaging and information campaigns surrounding the passage of the local mandates (Rees-Jones and Rozema 2023). The second indicator—which equals one in the regions of King County after May 2010, Albany County after September 2010, and the state of Vermont—evaluates whether the effect of calorie posting laws depends on the stringency of enforcement. As noted in Section 4, the likelihood that consumers observed calories on menus was lower in areas not actively enforcing local mandates.

The results are shown in column (2) of Table 3. Actively enforced laws lead to a 0.191 kg/m^2 reduction in BMI, a difference similar to the reduction of 0.174 kg/m^2 observed in column (1). The estimated effect on the indicator for passage of the local law indicates a lack of anticipatory effects; the effect is positive but near zero and imprecisely estimated. Evidence on differential effects by the stringency of enforcement is mixed. The coefficient for the variable indicating lack of active enforcement is 0.126 but is not statistically significant. However, the magnitude is large enough to render the total effect of calorie mandates in areas without active enforcement small (-0.06 kg/m^2) and statistically insignificant.

5.4. Checks of Model's Validity

Appendix B provides estimates from a series of robustness checks. As shown in Online Appendix Table B.2, the results are robust to (i) excluding all control variables ($\boldsymbol{X}_{ict}$ and $\boldsymbol{Z}_{ct}$); (ii) controlling for only basic demographic control variables (age, gender, and race/ethnicity); (iii) excluding county time-varying variables ($\boldsymbol{Z}_{ct}$); (iv) excluding county-specific time trends; (v) controlling for covariate-specific time trends, household income, employment status, calories consumed outside the home, food prices, smoking, exercise, and drinking; (vi) changing the sample period, (vii) excluding counties adjacent to the treated counties; (viii) excluding sparsely populated counties; (ix) using synthetic control methods to select the control group (Abadie et al. 2010); (x) clustering the standard errors at larger geographic units than counties; and (xi) using the Borusyak et al. (2024) estimator to address concerns about potential bias due to staggered treatment timing and treatment effect heterogeneity (Goodman-Bacon 2021). Online Appendix Table B.3 shows that the results are robust to excluding observations from one treated or control area at a time. Online Appendix Table B.4 shows that the estimated impact on BMI is largest for individuals who more frequently eat at restaurants affected by calorie posting laws. Finally, Online Appendix Table B.5 and B.6 report a series of falsification tests showing that the laws did not have "effects" in the years before they were implemented or cause unexpected changes in the control variables. These checks provide additional evidence that the estimates reflect the causal impact of calorie posting laws instead of the Great Recession or other potential confounds.

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To further check for differential pre-treatment trends and examine the evaluation of the treatment effect over time, we estimate event-study regressions. Specifically, we replace LAW_{ct} in equation (1) with a series of relative time variables indicating the lags and leads for whether the law in that particular county will be implemented in a given number of years (e.g. in the next 1–12 months is coded as −1, 13–24 months as −2, etc.) or has been implemented a given number of years ago (e.g. 0–11 months ago is coded as 0, 12–23 months ago as 1, 24–35 months as 2, etc.). These indicators are all set to zero for the control group.

Even though monthly or quarterly periods are technically feasible, we choose annual leads and lags for our preferred event study model for two reasons. First, more narrow bins would be based on fewer observations and therefore lead to noisier estimates. Second, the BRFSS data are only designed to be nationally representative and state representative for a given year. As different counties are surveyed in different months, bins narrower than a year will be unbalanced in the composition of the treatment group. As noted by Goodman-Bacon (2021), if the composition of the treatment group changes across event study bins, the estimated treatment effect will fluctuate across periods even if there is no true heterogeneity across treated units over time.[27] For the same reason, we accumulate indicators for beyond 14 years before the law's implementation ($t < -14$) into a single variable and the indicators for beyond 2 years after the law's implementation ($t > 1$) into another variable. Estimates for these indicators are not shown in the figures as they are outside the window in which all treated units are in the sample.

Following Miller (2023), we benchmark each estimate against the average of the pre-treatment coefficients, though the results are broadly similar using $t = -1$ as the reference period. Finally, we exclude county time trends as Wolfers (2006) makes a compelling case that including treatment-unit-specific time trends is inappropriate in this type of event-study specification.

Results from this event-study regression are shown in Figure 2. The event study results indicate a statistically significant relative decrease in BMI of 0.26 kg/m^2 during the first year after implementation of a local mandate and a relative decrease in BMI of 0.24 kg/m^2 ($p = 0.051$) during the second year. Thus, the impact of the calorie labeling laws is relatively constant over the first two years. Moreover, the estimated impact from the difference-in-differences design—a 0.17 kg/m^2 reduction in BMI—falls within the confidence interval for both post-treatment periods. Consistent with the identifying assumption, there is no obvious trend in BMI in years prior to the law's implementation, and only one of the fourteen pre-treatment coefficients is statistically

27. It is possible that even using annual bins does not fully solve the problem since these bins straddle years (e.g. for a law implemented in July 2009, 12 months prior is July 2008–June 2009), and representativeness is only assured for a given calendar year. However, the results remain very similar if we classify years before and after treatment as being based solely on calendar year (effectively ignoring treatment month).

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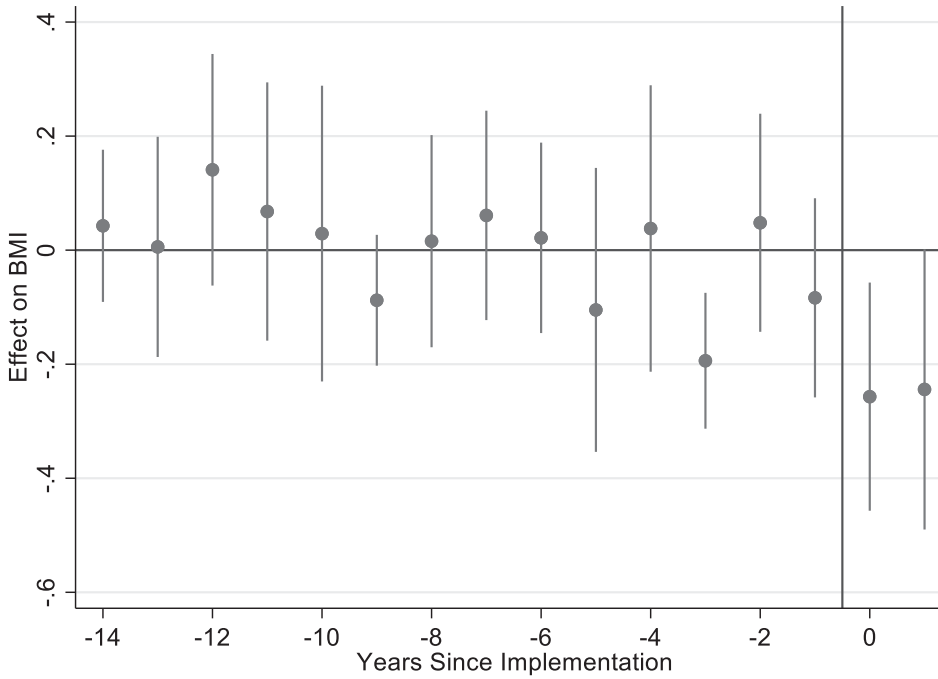

FIGURE 2. Event study results of calorie posting laws on body mass index. This figure displays point estimates and their 95% confidence intervals, which allow for clustering within county, for the lags and leads of the event-study regression. Pre-treatment indicators are accumulated beyond 14 years, and post-treatment indicators are accumulated beyond 2 years. Following Miller (2023), the reference period is the average of the coefficients in each period prior to the implementation of the calorie posting law. Observations are weighted using the BRFSS sampling weights. The control group consists of other counties in MSAs with at least one "treated" county, plus New England, but excludes any counties/states in these areas that passed but did not implement a calorie labeling law. Controls include binary indicators for the individual's sex, age, race/ethnicity, marital status, and education level, as well as county-level population density, median income, and unemployment rate. County and month∗year fixed effects are also included.

significant. Moreover, the estimated pre-treatment coefficients should reduce concerns that our estimates are driven by the Great Recession. We do not observe decreases in BMI during previous recessions, for example, the 2001 recession, which corresponds to estimates in the range of $t = -6$ to -9.[28,29]

28. Online Appendix Figure B.1 displays event study figures using quarterly bins. These estimates are much noisier due to the decreased sample size in each bin and compositional changes in the counties being surveyed in each quarter. Nonetheless, they show no consistent trend in the estimates prior to the implementation of the laws and many of the quarterly estimates for bins after the implementation of the laws are negative. Further, the average of the post-treatment estimates is relatively close to the main two-way fixed-effects estimates presented earlier.

29. In Online Appendix Figure B.2, we display results from event-study models with the control variables as outcomes. Results are discussed in the text of Online Appendix B.

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TABLE 4. Effects of chain restaurant calorie posting laws on smoking, drinking, and exercising.

	P (Smoker)	Drinks per month	*P* (any exercise)	Minutes per week moderate exercise	Minutes per week vigorous exercise
Law implemented	0.001	−0.167	−0.001	−3.283	−0.828
	(0.007)	(0.379)	(0.012)	(3.419)	(1.946)
Law implemented but no longer actively enforced	0.016	0.064	0.008	2.083	1.734
	(0.006)	(0.381)	(0.010)	(2.915)	(1.475)
Law passed but not yet implemented	−0.007	−1.037	0.012	−0.717	0.965
	(0.005)	(0.576)	(0.009)	(2.307)	(1.749)
Sample size	591,432	526,857	545,840	194,705	194,705
Mean of outcome	0.192	10.979	0.762	49.600	30.093

Notes: Results shown are average marginal effects from probit models for smoker and any exercise and negative binomial models for drinks per month and minutes of exercise per week. Standard errors, heteroskedasticity-robust and clustered by county, are in parentheses. Observations are weighted using the BRFSS sampling weights. The control group includes other counties in MSAs with at least one "treated" county, plus New England, but excludes any counties/states in these areas that passed but did not implement a calorie labeling law. The controls include dummies for the individual's sex, age, race/ethnicity, marital status, and education level, as well as county-level population density, median income, and unemployment rate. County and month∗year fixed effects are also included, as well as linear county-specific time trends.

5.5. Other Outcomes that Could Influence BMI

We next explore the behavioral mechanisms through which the effect on BMI occurs. To do so, we estimate the effect of calorie labeling mandates on smoking, drinking, and physical activity; compensatory behaviors that could influence BMI in the absence of changes in calories consumed. The results are shown in Table 4. Across all of these outcomes, the estimates of the impact of implementation of local calorie mandates are small in magnitude relative to the sample mean and not statistically significant. These results rule out concerns that observed changes in BMI reflect changes in caloric expenditure as opposed to changes in calories consumed.

6. Understanding Why Calorie Labels Matter

The prior section documents changes in average BMI that are consistent with our theoretical model. In this section, we test predictions from Proposition 2 of our model. In doing so, we aim to understand the relative importance of the various channels through which calorie labels could impact choice. We first describe our empirical strategy, which is based on proxies for the different consumer types. In Section 6.2, we examine the influence of calorie labeling laws on life satisfaction for all individuals and the different consumer types, while Section 6.3 presents evidence to support causality. Section 6.4 examines heterogeneity in the effect on BMI using quantile regression. In Section 6.5, we present evidence from Survey 1 that, for some individuals, the anticipated response to the national calorie labels mandate reflects moral costs whereas for others, the anticipated response reflects perceived reductions in informational and

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non-informational internalities. Section 6.6 resents evidence from a survey experiment showing that calorie labels influence proxies for moral costs.

6.1. Approach to Testing the Importance of Information and Moral Costs

Recall that our theoretical model outlines an empirical strategy to identify the relative importance of different channels. Both Type *A* and Type *B* individuals are susceptible to moral costs that can influence calories consumed and welfare. However, internalities only have discernible impact on the choices of Type *B* agents. Hence, if calorie labels work predominantly through reductions in internalities, the impact on average life satisfaction is unambiguously positive; there will be little to no impact on the choices of Type *A*s and a welfare enhancing reduction in overconsumption by Type *B*s. In contrast, if calories labels work predominantly through the moral costs channel, we would expect a reduction in life satisfaction for Type *A*s and an ambiguous impact on Type *B*s. Hence, the prediction for the population-wide average effect is ambiguous and depends on the relative share of each type and the strength of the competing effects.

Evidence within economics and psychology suggests the existence of moral costs and the possibility that some people can be susceptible to moral costs but not internalities, thereby providing a rationale for our characterization of Type *A* agents. Goldfarb et al. (2013) show that consumers demand higher amounts of calories when ordering online. Herman et al. (2003) review the relevant literature and show that the presence of others impacts eating behavior, particularly when social norms disapprove excessive eating. Ceylan et al. (2022) show that individuals experience more intense feelings of embarrassment when ordering high calorie meals in the presence of others, particularly in an environment with calorie labels. Choi et al. (2019) found that health-concerned individuals (a reasonable proxy for our Type *A* agent) are more likely to consume high calorie items when the calories are shown "ending in 99." Given that the health effects of a one-calorie change in consumption are effectively zero, such non-linear effects can only be rationalized through psychological effects triggered by calorie information.[30]

We do not directly observe any individual's type; however, we can draw upon prior literature to develop proxies for individuals who are more likely to have less accurate beliefs about calories and higher non-informational internalities. The review article of Wehling and Lusher (2017) documents that while individuals with BMI greater than or equal to 30 systematically underreport calories consumed in meals, there is no such pattern for individuals in healthy weight ranges.[31] Similarly, individuals

30. The importance of moral costs is also supported in prior work (Glaeser 2006; Loewenstein and O'Donoghue 2006) arguing that soft paternalism toward an activity creates a psychic tax that arises due to the nudge stigmatizing certain behaviors such as the consumption of high calorie foods. Loewenstein and O'Donoghue (2006) argue that this can foster a neurotic attitude whereby even legitimate activities—e.g. consumption of high calorie items by Type *A* agents—become associated with negative emotions.

31. While it is tempting to use additional demographics to define types, Wehling and Lusher (2017) show there is no consistent pattern of differences in beliefs across other demographic characteristics, for example, race, age, and gender.

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with obesity have been shown to demonstrate lower levels of self-control than the healthy BMI counterparts (Fan and Jin 2014). Higher BMI is also associated with higher impatience and dynamic inconsistency (Ikeda et al. 2010; Courtemanche et al. 2015a).[32] Given the association between BMI and behaviors reflective of informational and non-informational internalities, we use weight class as a proxy for unobserved type. Specifically, we use BMI in the healthy weight range to proxy for a Type *A* agent and BMI in the overweight/obese range to proxy for a Type *B* agent. These proxies are not meant to be exact, but instead merely introduce a sharp enough dichotomy to enable comparison.

To provide evidence on the relative importance of information/non-information internalities and moral costs as channels through which calorie labels influence choice, we estimate a modified version of equation (1) with life satisfaction as the dependent variable, paying particular attention to the subsamples of healthy weight individuals (Type *A*) and overweight/obese individuals (Type *B*). Since the sample period for life satisfaction (2005–1010) is much shorter than the period for BMI, we omit county-specific time trends in our baseline specification but explore the sensitivity of our findings to their inclusion in Online Appendix Table B.8.[33]

6.2. Effect of Calorie Labeling Laws on Life Satisfaction

In the first column of Table 5, we display the estimates for life satisfaction for the full sample. These results provide evidence that the average consumer is worse off following the implementation of the local calorie mandates; life satisfaction falls by 3.6% of a standard deviation, an effect that is statistically significant at the 10% level. Viewed through the lens of the Kimball and Willis (2023) framework, this suggests that the introduction of calorie labels is associated with lower levels of elation (or short-run happiness) that reflects anticipated reductions in lifetime utility—a pattern consistent with calorie labels triggering moral costs.

While the effects for the full sample are suggestive of the presence of moral costs, a sharper test of our theoretical predictions is based upon heterogeneity of the effects across types. The second and third columns of Table 5 provide results for this exercise. Column 2 reports the estimate for healthy weight individuals (our proxy for a Type *A*) and emphasizes that calorie labels impart moral costs. We find a statistically significant 10.5% of a standard deviation reduction in life satisfaction amongst treated individuals

32. The association of higher BMI with impatience has been confirmed by several reviews and meta-analyses (Amlung et al. 2016; Barlow et al. 2016; Bickel et al. 2021).

33. Note that our theoretical model does not suggest that changes in life satisfaction are proportionally related to changes in BMI. Calorie labeling could induce a large moral cost with a minimal reduction in BMI, or vice versa. Hence, we cannot assume away meaningful impact on life satisfaction in our setting even though the impacts on BMI were modest. Although outside the scope of our model, a calorie-labeling mandate may cause an individual to sort away from their preferred restaurant to one that is not required to post calories. By doing this, the individual is able to order a similar-calorie meal but forego the moral cost. However, eating at a less preferred restaurant leads to lower consumption utility. Hence, individual welfare falls but calories consumed and weight are unchanged.

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TABLE 5. Effects of chain restaurant calorie posting laws on life satisfaction.

	All	Healthy weight	Overweight or obese
Law implemented	−0.036	−0.105	0.009
	(0.020)	(0.037)	(0.022)
Observations	82,203	30,834	50,121
Mean of outcome	0.000	0.074	−0.042

Notes: Standard errors, heteroskedasticity-robust, and clustered by county, are in parentheses. Observations are weighted using the BRFSS sampling weights. The control group consists of other counties in MSAs with at least one "treated" county, excluding any counties/states in these areas that passed but did not implement a calorie labeling law. The DC MSA and New England are excluded because they did not have any labeling laws during the life satisfaction sample period. The controls include binary indicators for the individual's sex, age, race/ethnicity, marital status, and education level, as well as county-level population density, median income, and unemployment rate. County and month∗year fixed effects are also included. Healthy weight includes individuals with BMI greater than or equal to 18.5 and less than 25. Overweight or obese includes individuals with BMI greater than or equal to 25. All includes individuals with a healthy weight, overweight or obese, and underweight (BMI less than 18.5).

following the implementation of calorie-posting laws. In contrast, the estimate for overweight and obese (our proxy for a Type *B*) individuals is positive but small and not statistically significant.

6.3. Checks of Model's Validity

Because the sample ends in 2010 for life satisfaction, we do not examine the impact of calorie-posting laws on life satisfaction by the stringency of the enforcement of the local laws. However, in Online Appendix Table B.7, we augment the baseline specification to include an indicator that the law was passed, but not yet implemented. Estimates from this model suggest that results in Table 5 do not reflect anticipatory changes.

We further examine the robustness of the results in Online Appendix Table B. We use a similar set of robustness checks to those for BMI, with the exception that we do not vary the length of the pre-treatment period given the short panel length for life satisfaction. The results are stable across specifications that vary the control variables, the construction of the control group, and the level of clustering of the standard errors (Online Appendix Table B.8). We further show that the results are not driven by one specific local law (Online Appendix Table B.9) and are robust to using an ordered probit model to examine life satisfaction categories (Online Appendix Table B.10). Finally, we show that there are no placebo effects in the period before the laws were implemented (Online Appendix Table B.11). Overall, the robustness of the results suggests that the estimates are causal and do not reflect changes in economic conditions due to the recession or other public health initiatives.

Figure 3 presents results from separate event-study regressions for individuals who are of healthy weight and individuals who are overweight or obese. For individuals who are of healthy weight, life satisfaction decreases in the first year following the implementation of the laws and further decreases during the second year. For individuals who are overweight or obese, the estimates following the implementation

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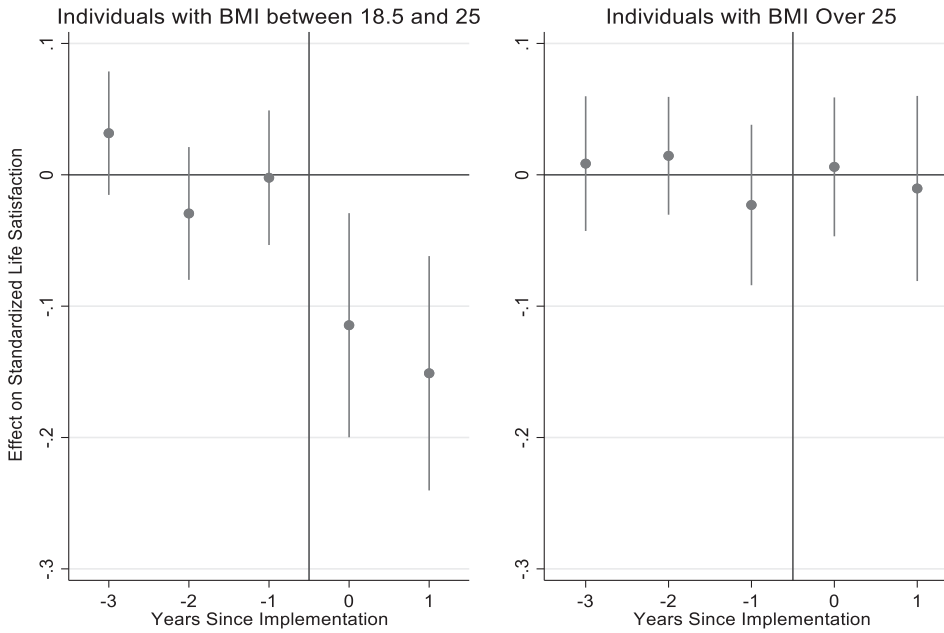

FIGURE 3. Event study results of calorie posting laws on life satisfaction. This figure displays point estimates and their 95% confidence intervals, which allow for clustering within county, for the lags and leads of the event-study regression. Pre-treatment indicators are accumulated beyond 3 years, and post-treatment indicators are accumulated beyond 2 years. The reference period is the average of the coefficients in each period prior to the implementation of the calorie posting law, following the recommendations of Miller (2023). In the regression, observations are weighted using the BRFSS sampling weights. The control group consists of other counties in MSAs with at least one "treated" county, excluding any counties/states in these areas that passed but did not implement a calorie labeling law. The DC MSA and New England are excluded because they did not have any labeling laws during the life satisfaction sample period. The controls include binary indicators for the individual's sex, age, race/ethnicity, marital status, and education level, as well as county-level population density, median income, and unemployment rate. County and month∗year fixed effects are also included.

of the laws are small and not statistically significant. Importantly, for both samples, the estimates for years prior to the implementation of calorie posting laws are not statistically significant and exhibit no clear trend; patterns consistent with the identifying assumption of both the difference-in-differences and event-study designs.[34]

Estimates of the impact of calorie-posting laws conditional on current BMI category may be biased if treatment leads to meaningful shifts in the distribution of respondents across weight categories. We therefore estimate ordered probit models analogous to equation (1) to determine whether calorie-posting requirements change

34. Online Appendix Figure B.3 displays the event study results using quarterly bins. The takeaways are the same as those discussed here, and the lack of any evident pre-treatment trend is based on more time periods.

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the probability that an individual is underweight, healthy weight, overweight, or obese. Results are presented inm Online Appendix Table B.12 and show that such effects are small in magnitude and not statistically significant.[35] Hence, it does not appear as if the proxies used to classify high or low internality types is a function of calorie-posting laws.

Another potential concern is that there may be differences in the likelihood that overweight and obese individuals observe calories on menus relative to healthy weight individuals. If so, our results could reflect differential exposure to calories rather than differences in the channels through which calorie labels impact choice across these types. To assess this concern, we examine data from Survey 1 on the likelihood that individuals of a given type reported that they noticed calories when eating at restaurants in the past 24-hours. We observe no meaningful differences; calories were noticed by 44.2% of healthy weight, 42.6% of overweight, and 41.2% of obese individuals.

It is thus unlikely that our results reflect endogenous sample selection or differential exposure to calorie counts across consumer types. Rather, our results suggest that calorie labels impact choice through multiple channels: the reduction of informational and non-informational internalities and increased moral costs. For Type *A* (low internality) individuals, we observe reductions in life satisfaction, which is consistent with the theoretical role of moral costs. For Type *B* (high internality) individuals, we observe no statistically discernible change in life satisfaction, suggesting reductions in the marginal internality but a compensating increase in moral costs.

6.4. Evidence on the Importance of Moral Costs from Quantile Regression Estimates

An alternate way to identify the relative importance of the different channels is to use quantile regression and examine effects across the BMI distribution. Our theoretical model predicts a reduction in BMI across all consumer types if calorie labels work predominantly through moral costs. In contrast, reductions in BMI should be concentrated amongst Type *B* agents if calorie labels work predominantly through a reduction in internalities. Testing for changes in BMI for individuals in the healthy weight (our proxy for Type *A*) region of the BMI distribution thus provides another strategy to identify impacts driven by moral costs.

Results from the quantile regression are shown in Figure 4 and highlight two important patterns. First, we observe statistically significant reductions in BMI for consumers in the 0.1 through 0.3 quantiles, which correspond to individuals who are of healthy weight. Weight loss in this range suggests that choices are influenced by moral costs and provides evidence consistent with observed changes in life satisfaction. Second, the largest point estimate occurs in the 0.9 quantile, which corresponds to

35. The weight loss corresponding to the average decrease in BMI shown in Table 3 is 1.5 pounds. The difference between the overweight and obesity thresholds of 5 BMI units corresponds to around 30 pounds at the sample mean height. Thus, given the small effect size on BMI, it is not surprising that the estimates of the impact of calorie-posting requirements on overweight and obesity are small in magnitude and not statistically significant.

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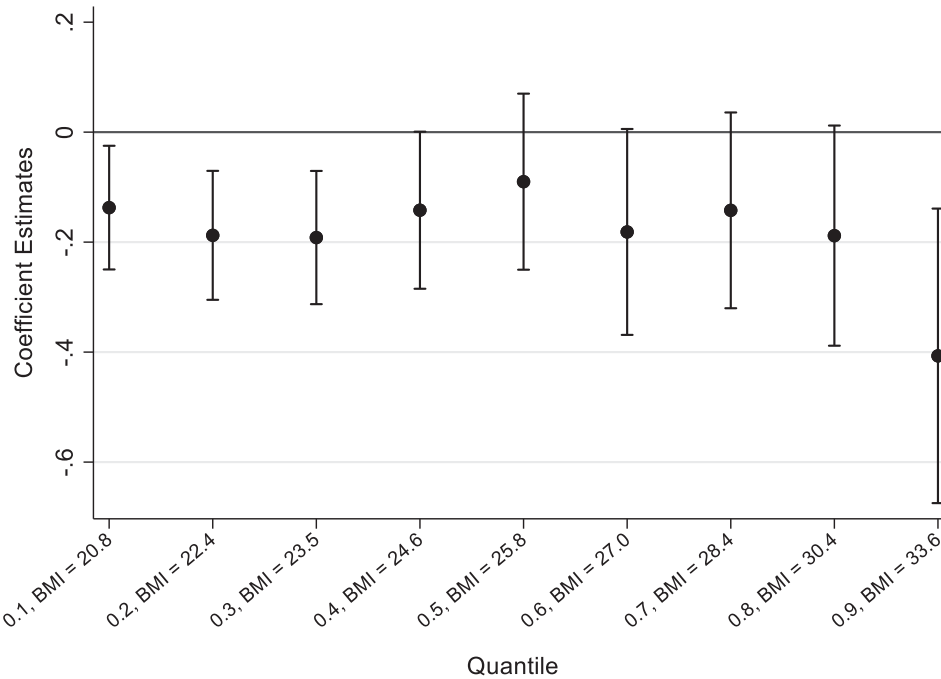

FIGURE 4. Quantile regression results. The figure displays point estimates and their 95% confidence intervals, derived from heteroskedasticity-robust standard errors. The sample size is 593,364 and observations are weighted using BRFSS sampling weights. The control group consists of other counties in MSAs with at least one "treated" county, plus New England, excluding any counties/states in these areas that passed but did not implement a calorie labeling law. The controls include dummies for the individual's sex, race/ethnicity, marital status, and education level, age, age squared, and county-level population density, median income, and unemployment rate. The model includes area and month∗year fixed effects. The *x*-axis displays the quantile, ranging from 0.1 to 0.9, and the BMI level at that quantile.

obese individuals with BMI $>$ 30; this quintile is a proxy for Type *B* agents whose choices could be affected through either internalities or moral costs. Thus, results from the quantile regression complement the findings for life satisfaction; moral costs and reduced internalities appear to both be important channels through which calorie labels influence choice.

6.5. Survey Responses about Restaurant Sorting

As a third method to disentangle the channels through which calorie labels impact choice, we draw upon the data from Survey 1. In the survey, we asked respondents the following question about the ACA:

> …this law will require restaurants with 20 or more locations to post calories, restaurants with fewer than 20 locations nationwide are exempt from the law and will not have to post calories. How do you think the law will affect the frequency with which you visit the restaurants required to post calories as opposed to those not affected?

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Respondents indicated whether they believed the frequency with which they would eat at chain restaurants would increase, decrease, or be unaffected.

We use the answers to identify lower bounds on the fraction of people who believe that calorie labels will trigger moral costs and those who believe that they will provide valuable information or other internality-related benefits. If calorie labels trigger moral costs, the utility of eating a meal at a chain restaurant would fall. For some of these individuals, this could change their preference ordering over restaurant types and a reduction in the frequency of eating at chain restaurants. Intuitively, such individuals will eat at an otherwise less preferred location to avoid moral costs triggered by calorie labels. In contrast, if calorie labels provide valuable information, the utility of eating a meal at a chain restaurant would rise. For some of these individuals, this could change their ranking over restaurant types and an increase in the frequency of eating at chain restaurants. In other words, such individuals will eat at an otherwise less preferred location to seek information from calorie labels.[36]

We define an individual influenced by moral costs as one who reports that the federal mandate will cause them to reduce the frequency of eating at chain restaurants. Based on responses from 20,123 individuals across four waves of the survey conducted between December 2014 and December 2017, approximately 7.2% of respondents expect to reduce the frequency of eating at chain restaurants.[37] We define an individual influenced by information as one who reports that the federal mandate will cause them to increase the frequency of eating at restaurants required to post calorie counts. Approximately 26.1% of respondents fall into this category. That non-trivial subsets of our sample report that they will either sort away from or toward chain restaurants provides additional evidence of moral costs and beneficial information, respectively.

6.6. Survey Experiment Suggests Importance of Moral Costs

To provide further evidence on the potential influence of moral costs, we designed a survey experiment (Survey 2) that was implemented in collaboration with Qualtrics.[38] Survey 2 was fielded in two waves in September and October 2022 and includes data from over 2,000 unique respondents. The survey experiment was designed to identify whether the presence of calorie information influenced psychological constructs—feelings of guilt and social appropriateness—that proxy for moral or psychological

36. The resulting impact on utility may be insufficient to trigger a change in where the person eats but instead impact what they eat and/or the utility received from meals consumed at chain restaurants. Using sorting to identify types of individuals thus provides a lower bound on the "true" proportion of such types in the population.

37. Out of the total, 40.3% of individuals report that the federal law will make them change what they purchase at chain restaurants.

38. Survey experiments are becoming an increasingly popular method to elicit beliefs about (or attitudes toward) various policies or actions and how they are shaped by informational nudges. Stantcheva (2023) and Haaland et al. (2023) provide excellent overviews on the design of survey experiments and informational interventions and their use in economics.

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costs. To establish a causal link between calorie information and stated feelings of guilt/appropriateness, we randomly assign respondents into information treatments that vary whether the description of a menu item includes calories.

Respondents were presented two types of questions designed to capture psychological costs: the first elicited feelings of guilt and the second an evaluation of the social appropriateness of a given action. To assess the guilt they would feel when ordering different menu items, subjects were asked to report: "*On a scale of 1 to 7, where 1 is "not at all guilty" and 7 is "very guilty," rate how guilty you would feel if you ordered the following menu items*." Respondents were provided, in random order, a description of six different meals—each of which cost $7.49—that include an entrée, side item, and drink.[39] Half of the respondents were randomly shown the caloric content of the items which ranged from 470 (Greek salad, apple, and bottled water) to 1,220 (roast beef sandwich with cheese, medium fries, and medium coca-cola).

We use data from the survey experiment to estimate the causal effect of calorie information on standardized measures of self-reported guilt in each meal separately. To increase the precision of the estimates, we report results from regressions that include control variables (age, gender, pregnancy and marital status, race/ethnicity, household size and income, educational attainment, and the month of the survey). Figure 5 plots coefficient estimates and 95% confidence intervals for the estimated treatment effect on standardized guilt for each meal.

The confidence intervals plotted in Figure 5 are ordered according to the number of calories in each of the given meals. As displayed in the figure, providing information on calories in a menu item increases the guilt from ordering high calorie items but has little to no impact on the guilt from ordering lower calorie items. Moreover, the presence of calorie labels increases the relative difference in the guilt associated with ordering the lowest and highest calorie menu items—there is a statistically significant difference in the estimated treatment effect between the lowest and highest caloric meals. These patterns are consistent with the presence of moral costs that are triggered when purchasing high-calorie items from a menu displaying calorie content.

Survey 2 included a second question that builds upon an approach developed by Krupka and Weber (2013) to elicit the social appropriateness of various actions to identify the role of norms and norm compliance on pro-social behavior. We adapt this approach to elicit the social appropriateness of ordering different drinks at a café and the extent to which such evaluations are shaped by calorie labels. Specifically, respondents were provided the following scenario:

> "Consider an individual who is taking an after lunch coffee break at a Starbucks Coffee Shop. The menu below gives a description of six different drinks of S size that the individual could order." Respondents were then assigned the following task: "For

39. The different meals were selected from large, national chain restaurants and reported the number of calories for each meal as reported on the restaurants' websites.

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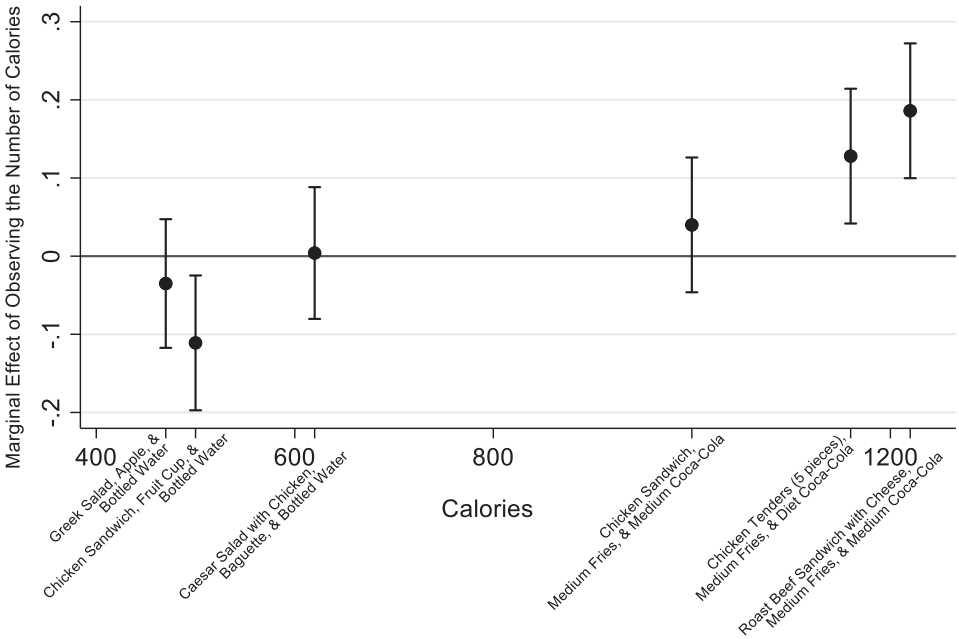

FIGURE 5. Estimates of calorie labels on guilt from ordering different menu items. This figure displays point estimates and their 95% confidence intervals, derived from heteroskedasticity-robust standard errors. The estimates are ordered along the *x*-axis according to the number of calories in the meal. The meals consist of an entrée, side item, and drink from large, national restaurant chains (two meals from Panera Bread Company, two meals from Chick-fil-A, and two meals from Arby's). Respondents rated how guilty they would feel from ordering each of the 6 meals, which were presented in a random order. The estimates represent the effect of observing the number of calories in the meal on the standardized measure of guilt. Control variables include age, gender, pregnancy status, race/ethnicity, marital status, household size, educational attainment, household income, and the month that the survey was completed. The sample consists of adults at least 18 years old throughout the USA. The sample size is 2,022.

> each of these drinks, please indicate whether you believe that ordering that option is very socially inappropriate, somewhat socially inappropriate, somewhat socially appropriate, very socially appropriate."

We varied two dimensions of the underlying choice set: (i) the size of each drink and (ii) whether the description of each drink included information on caloric content. The size S of the six drinks in any given scenario was either Tall (12 fl. Oz.) or Venti (20 fl. Oz.). Both the underlying size and the order of the set of six drinks was randomized at the respondent level. For any given choice set, we further randomized whether the description of each menu item included its caloric content such that half of the respondents observed menus with calorie information for each of the six items and half observe menus without such information.

We ordered the responses 1 through 4 such that a higher value indicates a more socially appropriate action and standardize the measure of social appropriateness for

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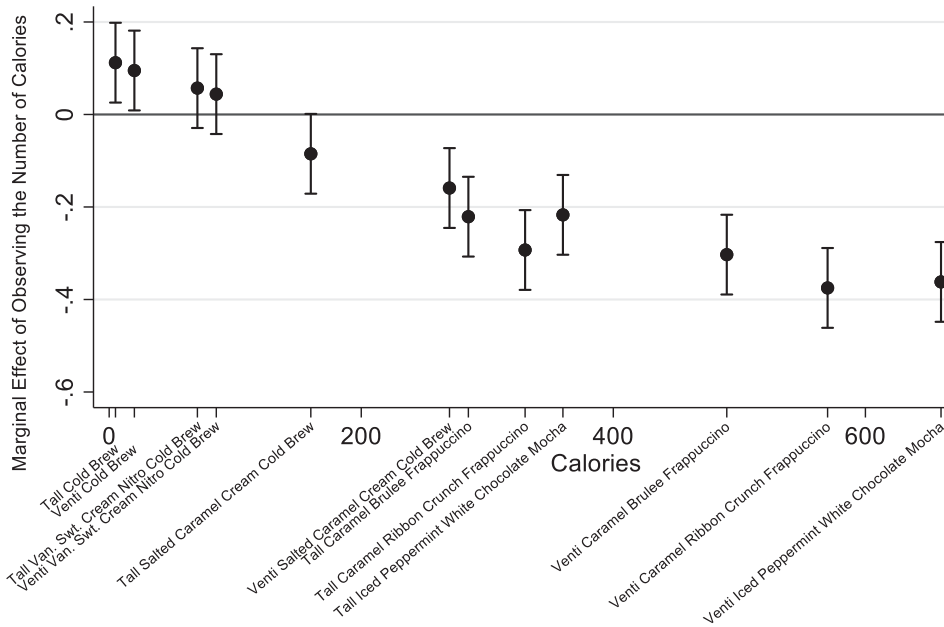

FIGURE 6. Estimates of calorie labels on social appropriateness for different menu items. This figure displays point estimates and their 95% confidence intervals, derived from heteroskedasticity-robust standard errors. The estimates are ordered along the x-axis according to the number of calories in the item. To better distinguish items along the *x*-axis, we present the number of calories in a Venti Cold Brew as 20 (instead of 5), the number of calories in a Venti Vanilla Sweet Cream Nitro Cold Brew as 85 (instead of 70), and the number of calories in a Tall Caramel Brulée Frappuccino as 285 (instead of 270). Respondents rated how socially appropriate it would be to order each of the 12 items which are available at Starbucks. Higher values indicate that ordering the drink is perceived as more socially appropriate. The estimates represent the effect of observing calorie information on the standardized social appropriateness rating. Control variables include age, gender, pregnancy status, race/ethnicity, marital status, household size, educational attainment, household income, and the month that the survey was completed. The sample consists of adults at least 18 years old throughout the USA. The sample size is 2,022.

each menu item. We estimate social appropriateness for each menu item as a function of an indicator for observing the number of calories in the drink and include controls to increase the precision of the estimates. Figure 6 plots the estimated treatment effect and 95% confidence intervals for each of the twelve different menu items. The figure is arranged according to the number of calories in each item.

As shown in the figure, observing calories impacts social appropriateness ratings. For the lowest calorie items, perceptions of the appropriateness of ordering the drink are greater in the presence of calorie labels. We observe the opposite effect for the highest calorie items; the estimated treatment effects are negative and statistically significant suggesting that perceptions of the appropriateness of ordering such drinks decrease when participants observe calorie labels. As such, calorie labels serve to

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increase the difference in the perceived appropriateness of ordering the lowest and highest calorie drinks.

Overall, results from the survey-based experiment show that calorie information influences both feelings of guilt and social appropriateness. Estimated treatment effects are more pronounced, and of opposite sign, for higher calorie items. Data from Survey 2 provide supplemental evidence that moral costs are a channel through which calorie labeling laws influence choice.

7. Conclusion

To provide information to consumers at the point of purchase, policymakers have introduced laws requiring chain restaurants to post calories on menus. Prior research exploring such laws has focused on identifying the impact on calories consumed and/or BMI. In this paper, we examine *why* such laws influence consumer choice. We develop a theoretical model through which the information on restaurant menus can influence choice through both reduced internalities and increased moral costs. While our model shows that the introduction of calorie labeling laws will lead to a reduction in calories consumed, the impact on welfare is ambiguous. Yet, predictions of the model suggest an empirical strategy to uncover the presence and importance of each channel.

We test these predictions by combining BRFSS data on BMI and life satisfaction with other secondary data sources. We explore the impact of city, county, and state calorie posting requirements on BMI and life satisfaction by comparing changes in these outcomes over time in geographic areas that implemented such requirements to changes in the surrounding geographic areas using both difference-in-differences and event-study style designs. Results from our empirical analysis show that calorie-posting laws lead to small reductions in average BMI; a change that reflects both increased moral costs and the salience of information.

The importance of moral costs is identified through several pieces of evidence: weight loss and reductions in life satisfaction for individuals in the healthy weight range of BMI, survey data showing that a notable fraction of individuals anticipate taking costly action to avoid calorie counts, and a survey experiment showing that proxies for moral costs—feelings of guilt and social appropriateness ratings—are impacted by the presence of calorie labels. The importance of informational and/or non-informational internalities is inferred from two sources: reductions in BMI without change in life satisfaction amongst overweight and obese individuals and survey data showing that other individuals anticipate taking costly action to seek information on calorie counts.

The estimated 0.17 kg/m^2 reduction in average BMI suggests that calorie-posting requirements are unlikely to have meaningful impacts on obesity-related health conditions. From a policy perspective, the observed effect sizes are lower than what would be predicted given evidence from field experiments testing the impact of calorie labels on calorie choice. Yet, such divergence is not surprising given recent work on scalability. We provide evidence that some consumers are willing to change where they

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eat to avoid exposure to calorie labels and many establishments choose not to comply with the law. These are predictable threats to scale that likely attenuated the observed impact of labeling laws relative to what would be inferred from field experiments that do not allow for such sorting or imperfect compliance. In this regard, our results highlight the need to complement evidence-based policy, or efficacy tests, with policy-based evidence, or Option C thinking (List 2024).

Our results document heterogeneous welfare effects that include reductions in well-being for healthy weight individuals. We are not the first to demonstrate heterogeneous welfare effects for "nudges" and policies motivated by behavioral insights (e.g. Handel 2013; Allcott and Kessler 2019). One should not use the potential for welfare losses by certain types to rule out the use of calorie labels or similar policies. Rather, it is important to better understand their benefits and costs and tailor nudges to subsets of the population most likely to benefit (Allcott and Kessler 2019).

Are there alternate models that could explain our empirical findings? Any such model must work along multiple channels that differentially impact healthy and overweight individuals. The first must cause individuals whose BMI is in the healthy weight range to lower calorie intake and experience a reduction in life satisfaction. The second channel must cause individuals with high BMI to lower consumption without impacting life satisfaction. This rules out any explanation with a single driver of behavior change such as imperfect information. More generally, our data are at odds with *any* variant of a neoclassical model whereby the introduction of calorie labeling laws leads individuals to (i) consume less and experience higher utility, (ii) avoid chain restaurants and eat at home or in restaurants that are not required to post such information, or (iii) not change consumption patterns but get less utility when dining at a chain restaurant.

Another model that is inconsistent with our results is one in which consumers are fully rational but have imperfect information regarding calorie counts. In such a setting, the introduction of calorie labels could lead the decision-maker to reoptimize and consume fewer calories. However, any such change would lead to a higher level of utility—a prediction that is at odds with our finding of weight loss and lower life satisfaction for healthy weight individuals. Moreover, reductions in calories consumed and subsequent BMI in such setting can only arise if all agents were to systematically underestimate the caloric content of restaurant meals.

Future work should focus on the welfare effects of calorie labels in greater detail. We envision three avenues for such work. First, it is important to directly estimate welfare effects as opposed to inferring the direction of welfare change using information on stated well-being and proxies for moral costs. Second, it is important to extend welfare calculations to incorporate changes in externalities related to obesity. Third, researchers should explore ways to mitigate the adverse effects of "nudges" without compromising their overall impact on the targeted behaviors.

An additional direction for future work is to incorporate dynamics and adaptation. Our results should be considered as short run estimates necessary for testing our theory, which is a static model. Further work should extend our theory to consider a longer time-horizon to allow for adaptation and the derivation of long-run welfare effects.

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Supplementary data

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